

Introduction to TACC Computing Cluster



Scalable Computational Intelligence
Texas Advanced Computing Center (TACC)

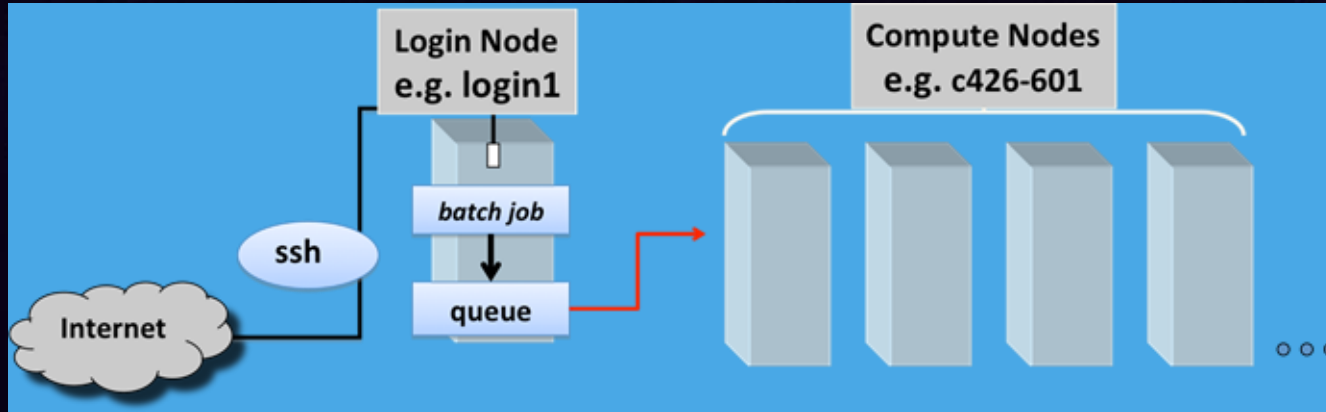
Overview

Practical introduction to using TACC resources

- Nodes and File system at TACC
- How to Remotely Access TACC
- Navigating the File System on TACC machines
- Hands On: Accessing TACC via TAP



Login vs Compute Nodes



Login Nodes

- Shared by multiple users
- Not meant for CPU or memory intensive jobs
- Meant for managing file and compiling code

Compute Nodes

- Where research computation occurs
- Training machine learning models, running simulations, etc.

Types of Compute Nodes: Partitions

- Compute nodes are divided into different pools called queues
 - For example here are a few queues on Frontera
 - development (interactive sessions)
 - small (one or two node jobs)
 - normal (≥ 3 node jobs)
 - rtx (gpu)
 - rtx-dev (interactive sessions with gpus)
 - nvdimm (large memory nodes)
 - Refer to the user guide for the queue and accounting details



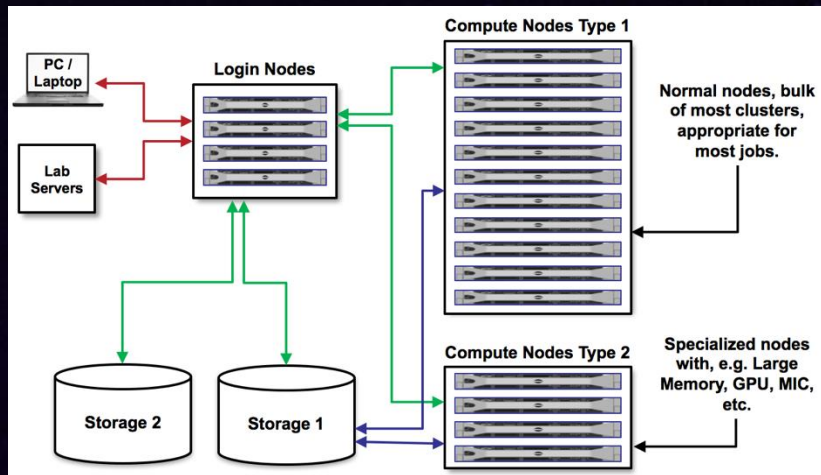
File Systems

Three (3) main file systems

- **\$HOME**
 - Quota (25G)
 - Backed up
 - Used for: small scripts, environment settings, and other routine file management task
- **\$WORK**
 - Quota (1TB)
 - Not Backed up
 - Used For: Intermediate staging/preparation of files, storing original datasets
- **\$\$SCRATCH**
 - No Quota
 - Not Backed up, subject to purge
 - Used for: High I/O bandwidth jobs

Refer to Frontera User Guide / TACC tutorials

- <https://frontera-portal.tacc.utexas.edu/user-guide/files/>
- <https://docs.tacc.utexas.edu/tutorials/managingio/>

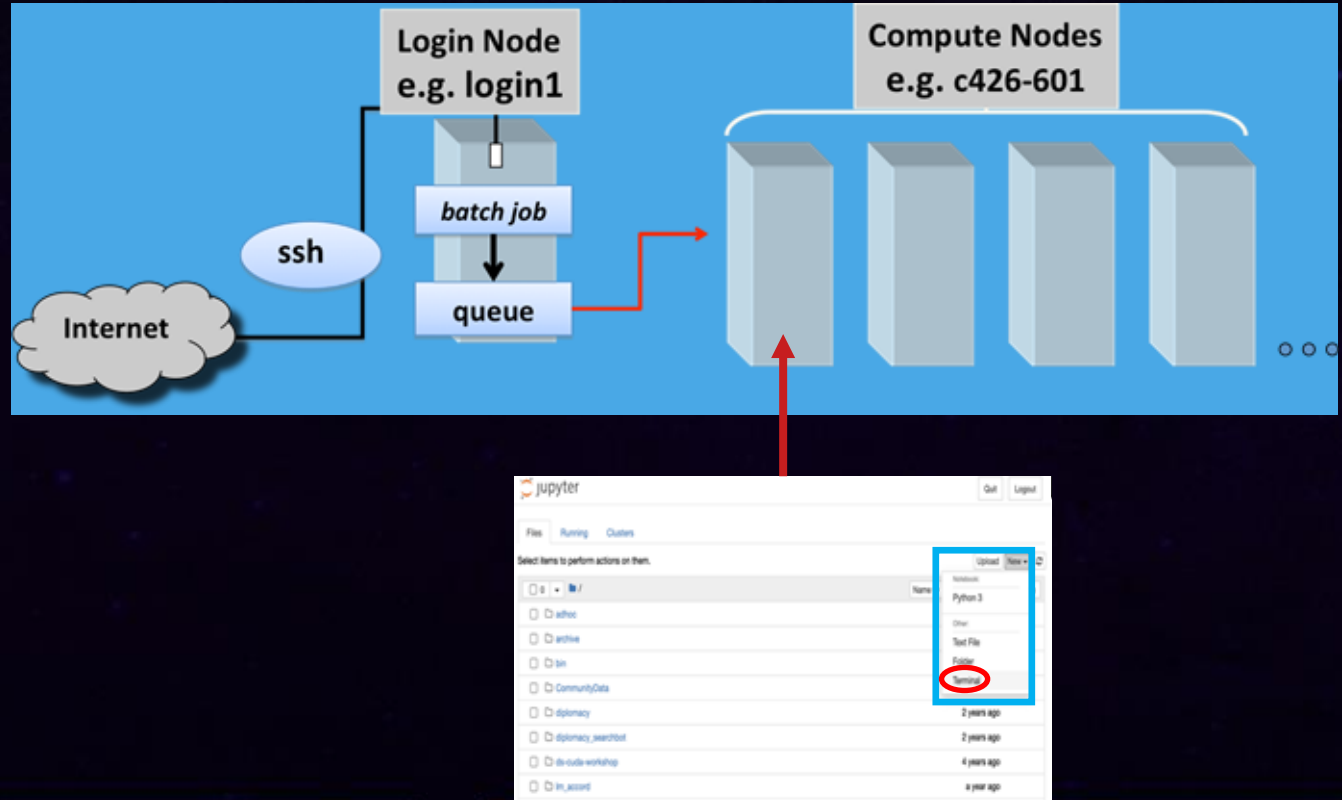


- **/tmp**
 - Each compute node comes with local storage (/tmp)
 - Fast I/O
 - **Temporary:** storage only last the duration of the job and should be copied for permanent storage

How to Remotely Access TACC

We will remotely access Frontera in two ways in this event:

1. SSH: Access login nodes via command line
2. TACC Analysis Portal: UI for accessing jupyter notebook session on a compute nodes



Accessing Compute Nodes

Let's assume we have logged into TACC via ssh and we are on a login node. How do we connect to a compute node?

- submit a batch job via the sbatch command
 - job waits in queue until resources are available
- activate an interactive session via **idev**
 - idev
 - Allows you to get a compute node for testing purposes
 - -m option lets you specify time in minutes
 - idev -m 30



Bash – basic file commands

- TACC machines use the Linux operating system
- Default shell and most popular is bash (Bourne Again Shell)

Command	Function
<code>pwd</code>	Shows absolute path of your present working directory
<code>ls <directory name></code>	Lists a directory (if mentioned) or present working directory
<code>mkdir <name></code>	Creates a new directory with mentioned name
<code>cd <name></code>	Changes working directory to name mentioned (or go to home directory if nothing is specified)
<code>cp <File1> <File2></code> <code>cp <File1> <Directory></code>	Makes a copy of File1 and names it File2 or makes a copy of the file with the same name in Directory1
<code>cp -r <directory1> <directory2></code>	Makes a copy of directory1 and all its contents into directory2
<code>mv <File1> <Directory></code> <code>mv <File1> <File2></code>	moves File1 to Directory1 or rename File1 to File2
<code>rm <File1></code> <code>rm -r <Directory1></code>	Removes a file or Remove a directory (add -f flag to force remove...be careful)



module

- On TACC Systems the module system helps setup or teardown useful packages that are supported by TACC, on the fly
- To find more information on a module
 - `module spider package-name`
- To load a module
 - `module load package-name`
 - `module load package-name/version.number`

Check out the User guide to learn more:

https://lmod.readthedocs.io/en/latest/010_user.html



Hands on: Accessing TACC via the TACC Analysis Portal and Environment Setup

Development Environment



- ▶ Jupyter Notebook
 - ▶ <http://jupyter.org/>
 - ▶ Web application for creating and sharing computational documents.
 - ▶ It offers a simple, streamlined, document-centric experience.
 - ▶ Provides an interactive interface in the browser
 - ▶ Can view output and plots inline
 - ▶ Makes Iterative/Interactive Development easy
 - ▶ Can be turned into a pure script for non-interactive work

Setting up for the tutorial

- ▶ Note, the following setup only has to be done once.
- ▶ This is copying files into the right places so you can run these notebooks with us.

One-Time Setup: Course Materials & Environment

Run our automated setup script to get everything ready for the tutorial.

- Step 1: Log in via the Command Line

Open your Terminal (Mac) or Command Prompt (Windows).
SSH into Vista with the command
(Please replace <username> with your actual TACC account username)

```
ssh <username>@vista.tacc.utexas.edu
```

- Step 2: Run the Installation Script

Copy, paste the command below and press Enter:

```
bash /scratch/01596/jrduncan/spring_26_tutorial_install/install.sh
```

```
~ % ssh @vista.tacc.utexas.edu
** WARNING: connection is not using a post-quantum key exchange algorithm.
** This session may be vulnerable to "store now, decrypt later" attacks.
** The server may need to be upgraded. See https://openssh.com/pq.html
To access the system:

1) If not using ssh-keys, please enter your TACC password at the password prompt
2) At the TACC Token prompt, enter your 6-digit code followed by <return>.

[If you are facing issues logging in, please use our login wizard at
https://accounts.tacc.utexas.edu/login_support to troubleshoot.]

( @vista.tacc.utexas.edu) Password:
( @vista.tacc.utexas.edu) TACC Token Code:
Last login: Thu Mar 26 14:29:17 2026 from 128.62.183.218
```

What is this script doing

- Copies Files:** It automatically moves all the notebooks and datasets for our tutorial directly into your personal \$HOME directory.
- Loads Software:** It configures your account to load Python so you don't have to do it manually.
- Preps Jupyter:** It registers your Jupyter kernel so it will run smoothly on the TACC Analysis Portal (TAP).

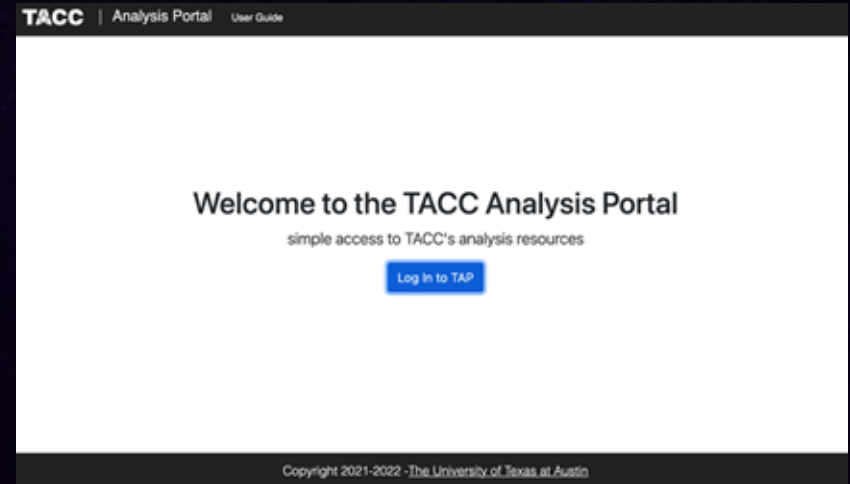
One-Time Setup

After running this enter **1s** and make sure you see an entry for **institute_summer_25**.

Once the script finishes, type exit and press **Enter**. You are now ready to open your web browser!.

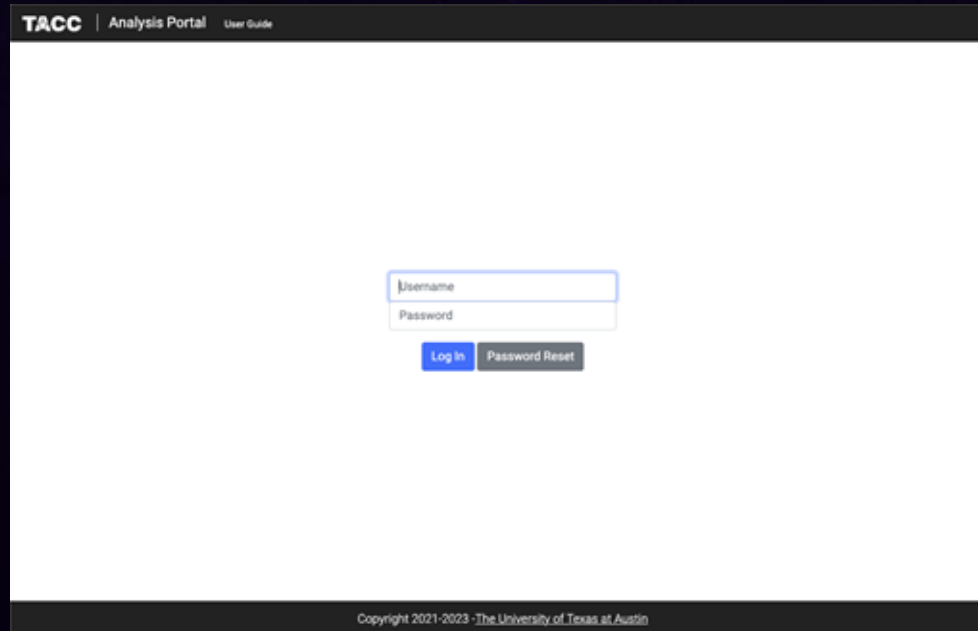
TACC Analysis Portal

- TACC Analysis Portal (TAP) provides access to interactive sessions on TACC machines
- Allows user to utilize web-based interactions on TACC computers
 - e.g. Jupyter notebook, Rstudio, remote linux desktop
- <https://tap.tacc.utexas.edu>



Login with your TACC account

- TACC Analysis Portal:
login with your TACC account credentials



The screenshot shows the TACC Analysis Portal login interface. At the top, a dark header bar contains the TACC logo, the text "Analysis Portal", and a link to the "User Guide". The main content area is white and features a login form with two input fields: "Username" and "Password". Below these fields are two buttons: a blue "Log In" button and a grey "Password Reset" button. At the bottom of the page, a dark footer bar displays the copyright notice: "Copyright 2021-2023 - The University of Texas at Austin".

Jupyter notebook:

TACC | Analysis Portal | [User Guide](#) jrduncan Log Out

Submit New Job

System

Application

Project

Queue

Nodes Tasks

Options

Job Name

Time Limit

Reservation

VNC Desktop Resolution

Submit Utilities

System Status

System	Status	Utilization	Job Count
Frontera Details	✓ Open	90%	Running: 290 Waiting: 20
Lonestar6 Details	✓ Open	100%	Running: 382 Waiting: 427
Stampede3 Details	✓ Open	89%	Running: 335 Waiting: 818
Vista Details	✓ Open	100%	Running: 197 Waiting: 134

Past Jobs

DCV-Vista	03/20/2026	Details	Resubmit
DCV-Vista	03/20/2026	Details	Resubmit
JNB-Vista	03/20/2026	Details	Resubmit
JNB-Vista	02/04/2026	Details	Resubmit

TACC

TEXAS

Waiting for Job to start (can take a minute)

TAP Job Status

Job: Jupyter notebook on Vista (668071, 2026-04-16T09:20-05:00)

Status: PENDING

Refresh: in 55 seconds

Message:

TAP: Your script has been submitted to Vista but it is not yet running.

`sinfo -p gh-dev` output is:

PARTITION	AVAIL	NODES(A/I)
gh-dev*	up	19/1

Check Status

End Job

Show Output

Back to Jobs

Once job is running you should see the following. Click Connect to launch the jupyter notebook

TAP Job Status

Job: Jupyter notebook on Vista (631103, 2026-03-20T14:43-05:00)
Status: RUNNING
Start: 03/20/2026 14:43
End: 03/20/2026 16:43
Refresh: in 841 seconds

Message:

TAP: Your session is running at https://vista.tacc.utexas.edu:60571/?token=7u3r0NcGJrhLkubixlvPp80_IHlpFVj2VdtBv656u_g

Connect

Job

Show Output

Back to Jobs

Jupyter notebook: Accessing New Python Notebook

The screenshot shows the Jupyter web interface. At the top, there's a header with the Jupyter logo and the word "jupyter". Below this, there are tabs for "Files", "Running", and "Clusters". The "Files" tab is selected. A message says "Select items to perform actions on them." Below this is a file browser showing a list of files and folders. The folder "institute_summer_25" is highlighted with a red circle. A blue arrow points from this folder to a larger red circle on the right containing the text "Institute_summer_25".

jupyter

Files Running Clusters

Select items to perform actions on them.

0 /

- ☐ CosmicAI
- ☐ course_containers
- ☐ cvw_cnn_install_script
- ☐ hold
- ☐ install_miniforge_cuda
- ☐ **institute_summer_25**
- ☐ old_educational
- ☐ pymgrid2_notebook
- ☐ sci_tacc_education_materials
- ☐ install
- ☐ jupyter.out

Institute_summer_25

Jupyter Notebook: Open Existing File

- Click on the [institute_summer_25](#) directory in the list
- Click on **KNNs** directory in the list
- Click on the file notebook file [intro_sklearn.ipynb](#)
- A Jupyter notebook should launch in a new tab
- Check to make sure you can run this notebook in the containerized kernel



Thanks!!

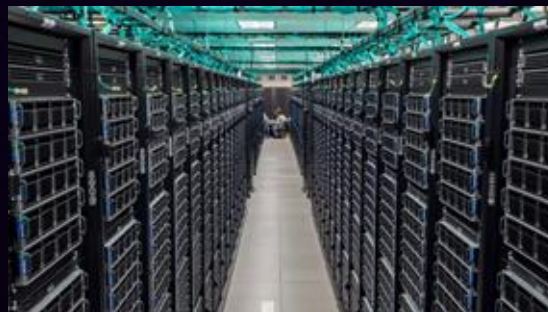
- Any Issues on TACC Systems that we can not help you with?
 - Open a ticket at <https://tacc.utexas.edu/about/help/>



Supplemental Slides

File Transfer Mechanisms

- Copying files from your local machine to TACC system
- Mac/OSX or Linux
 - Open Terminal window
 - Use scp (Secure Shell Copy)
 - Copies over ssh connection
 - -r option for recursive



```
scp <local file name> user@frontera.tacc.utexas.edu
```

```
scp -r <local folder name> user@frontera.tacc.utexas.edu:
```



File Transfer Mechanisms (2)

- Copying files from your local machine to TACC system
- Windows
 - Download WinSCP (<https://winscp.net>)
 - Double click installer
 - Choose “Explorer” Option
 - Hostname: `frontera.tacc.utexas.edu`
 - Username: `username`
 - TACC Code: <Use Code from your credentials handout>
- Drag and Drop Files/Folders to copy to Stampede2



Command Line Editing

- Often will be required
 - Working with code
 - Editing configuration files
 - Writing Bash Scripts
 - TACC Systems: Writing job submission scripts
- Vim
 - Vi Improved
- Emacs
- Nano
 - Simple, Easy to use



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 - `module load package-name/version.number`
- [Learn more here](#)

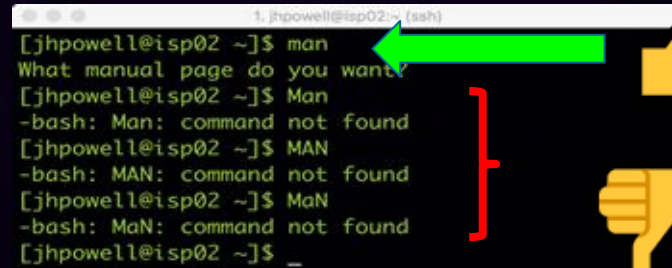


Linux/bash

- TACC machines use the Linux operating system
- Default shell and most popular is bash (Bourne Again Shell)
- Bash features auto completion of commands with tab key
- Bash commands are case sensitive

Example:
The “**man**”
command

`man` $\neq$ `Man` $\neq$ `MAN`



A terminal window titled '1. jhpowell@isp02:~ (ssh)' showing the following commands and outputs:

```
[jhpowell@isp02 ~]$ man  
What manual page do you want?  
[jhpowell@isp02 ~]$ Man  
-bash: Man: command not found  
[jhpowell@isp02 ~]$ MAN  
-bash: MAN: command not found  
[jhpowell@isp02 ~]$ MaN  
-bash: MaN: command not found  
[jhpowell@isp02 ~]$ _
```

A green arrow points to the first line, and a red bracket groups the last three lines. A thumbs up icon is next to the first line, and a thumbs down icon is next to the last three lines.